Learning Objectives - Administer Network Traffic

- Introduction to Azure Load Balancer
- Introduction to Application Gateway
- Introduction to Network Watcher
- Lab 06 – Implement Traffic Management

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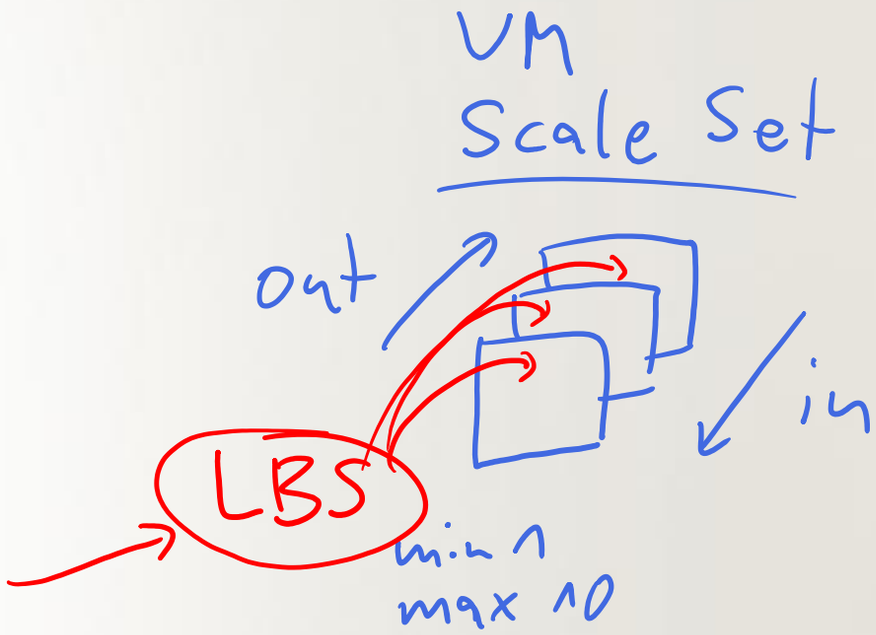

Introduction to Azure Load Balancer

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Learning Objective – Introduction to Azure Load Balancer

- Choose a Load Balancer Solution
- Implement a Public Load Balancer
- Implement an Internal Load Balancer
- Determine Load Balancer SKUs — Basic, Standard
- Create Load Balancer Rules
- Demonstration – Configure a load balancer
- Learning Recap

Exam objectives

Configure name resolution and load balancing

- Configure an internal or public load balancer
- Troubleshoot load balancing

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HTTPS

Choose a Load Balancer Solution

CDN

DNS

Feature	Application Gateway	Front Door	Load Balancer	Traffic Manager
Usage	Optimize delivery from application server farms while increasing application security with web application firewall.	Scalable, security-enhanced delivery point for global, micro service-based web applications.	Balance inbound and outbound connections and requests to your applications or server endpoints. VM, VMSS,	Distribute traffic to services across global Azure regions, while providing high availability and responsiveness.
Protocols	HTTP, HTTPS, HTTP2	HTTP, HTTPS, HTTP2	TCP, UDP	Any
Private (regional)	Yes	No	Yes	
Global		Yes		Yes
Env	Azure, non-Azure cloud, on premises	Azure, non-Azure cloud, on premises	Azure	Azure, non-Azure cloud, on premises
Security	WAF	WAF, NSG	NSG	

VM VM

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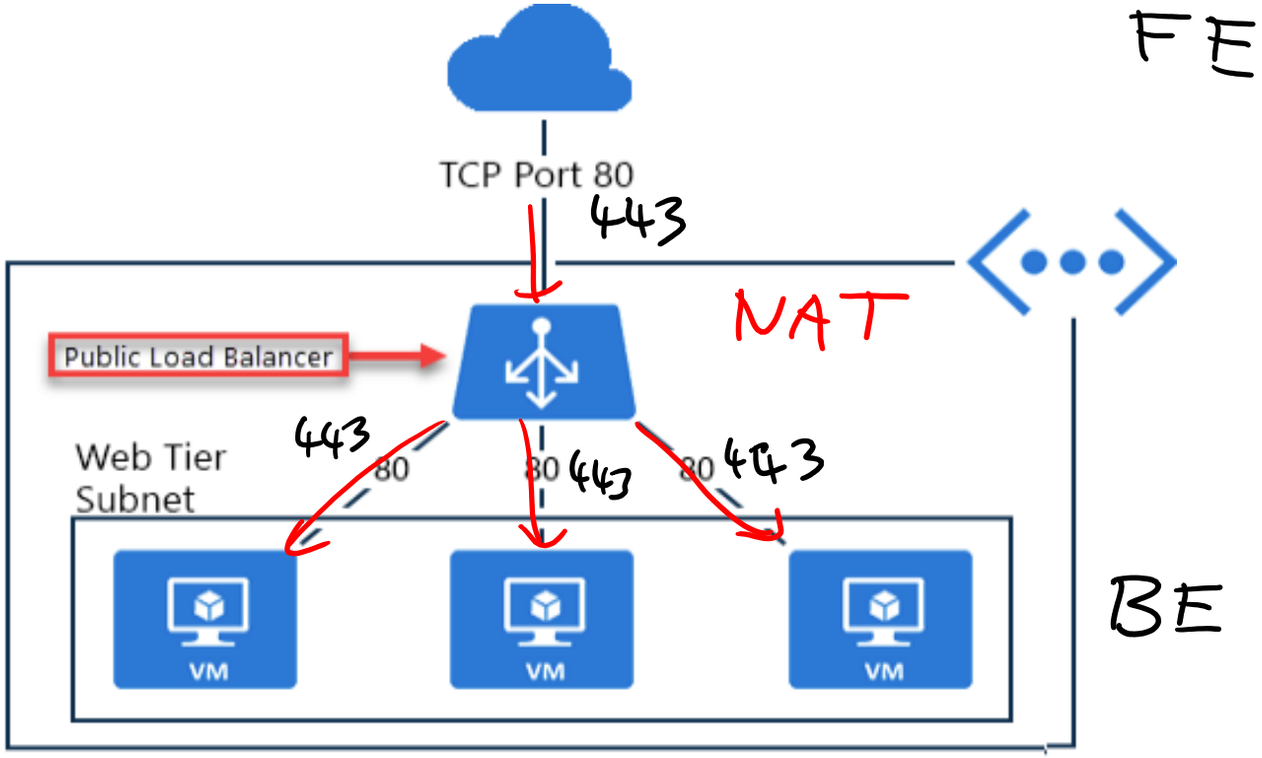
Implement a Public Load Balancer

- Maps public IP addresses and port number of incoming traffic to the VM's private IP address and port number, and vice versa
- Apply load balancing rules to distribute traffic across VMs or services



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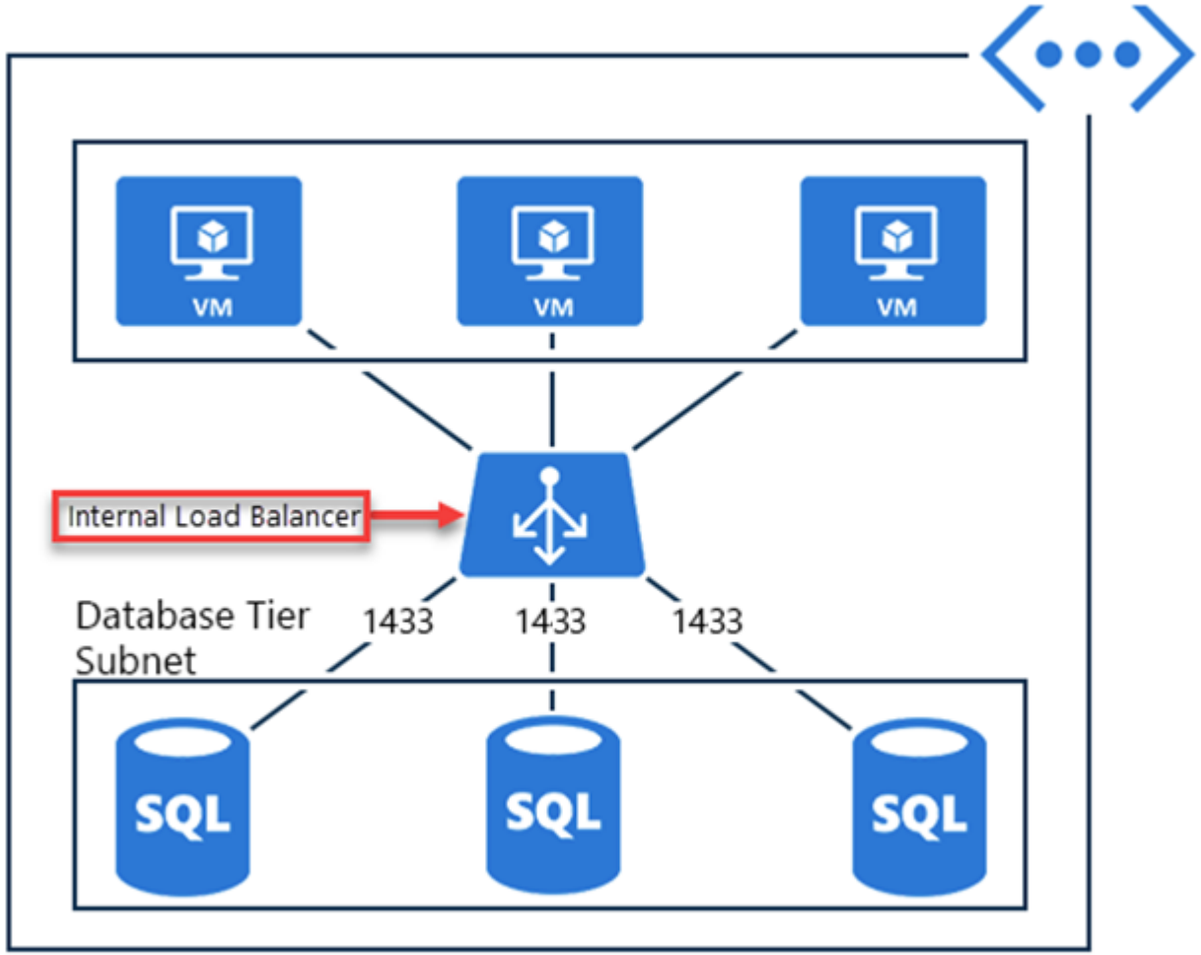
Implement an Internal Load Balancer

Directs traffic only to resources inside a virtual network or that use a VPN to access Azure infrastructure

Frontend IP addresses and virtual networks are never directly exposed to an internet endpoint

Enables load balancing within a virtual network, for cross-premises virtual networks, for multi-tier applications, and for line-of-business applications



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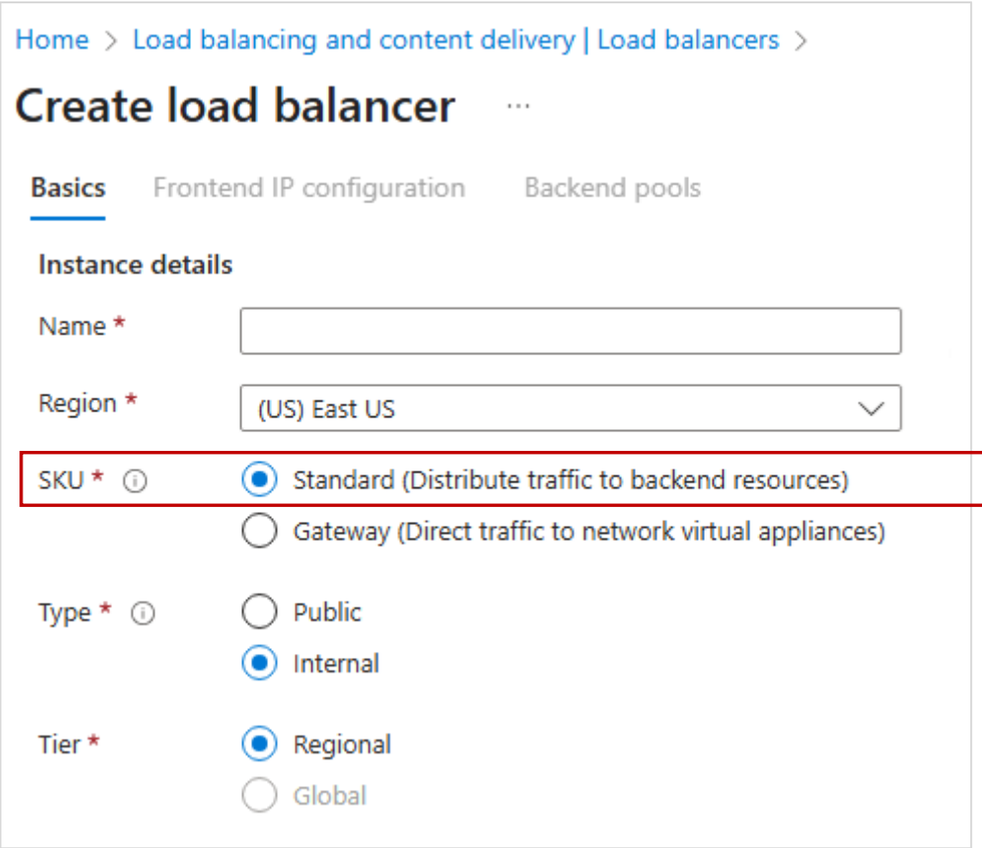
Determine Load Balancer SKU



Feature	Standard Load Balancer
Scenario	High performance, ultra-low latency, and high resiliency
Type	Internal or public
Backend type	IP based or NIC based
Multiple frontends	Inbound and outbound
Health probes	TCP, HTTP, HTTPS
SLA	99.99%

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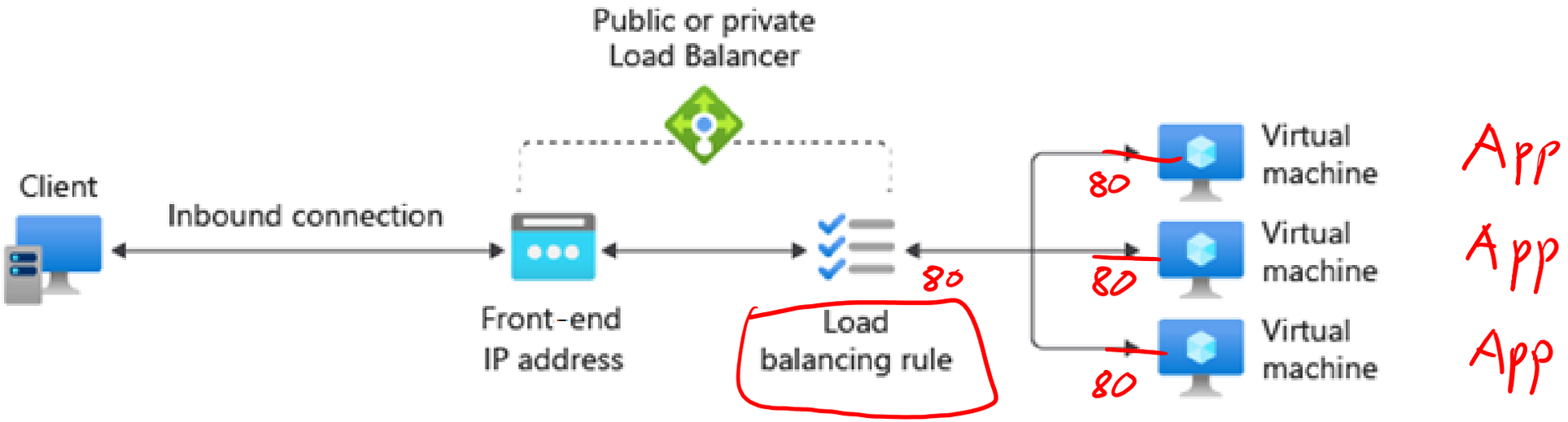
Create load balancer rules

Stateless



Maps a frontend IP and port combination to a set of backend pool and port combination

Rules can be combined with NAT rules

A NAT rule is explicitly attached to a VM (or network interface) to complete the path to the target

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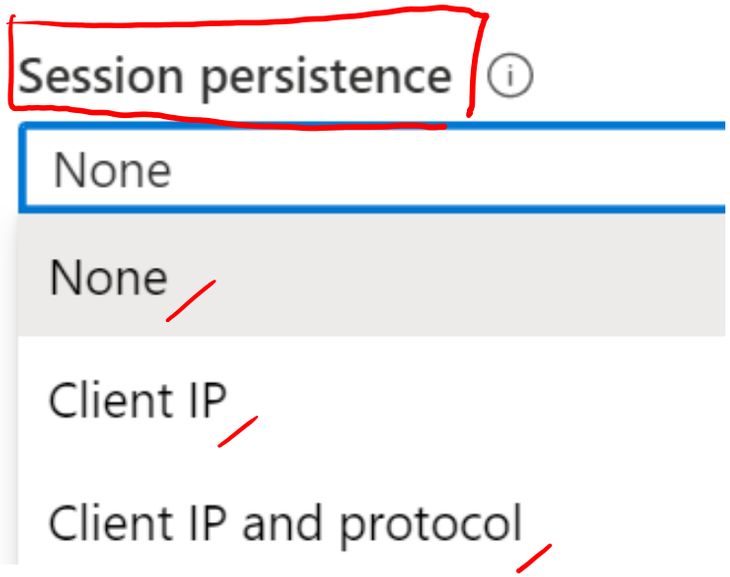
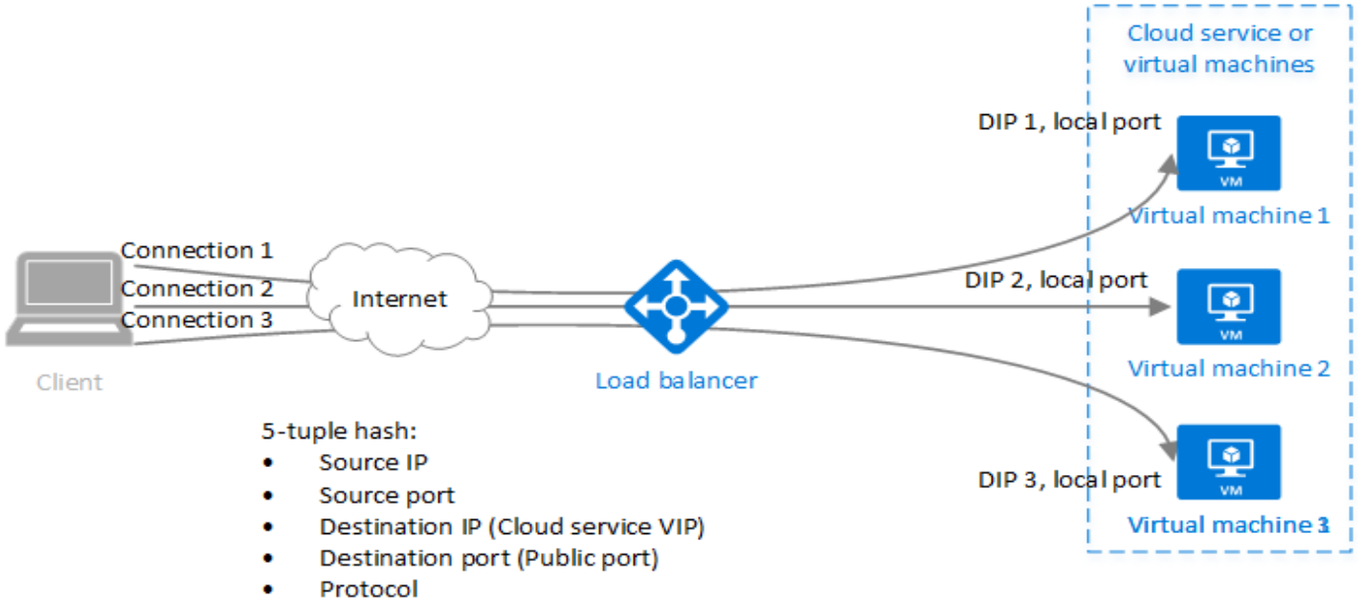
Configure Session Persistence (optional)





Session persistence specifies how client traffic is handled

None (default) requests can be handled by any virtual machine

Client IP requests will be handled by the same virtual machine

Client IP and protocol specifies that successive requests from the same address and protocol will be handled by the same virtual machine

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Introduction to Azure Application Gateway

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Learning Objective – Introduction to Azure Application Gateway

- Implement Application Gateway
- Determine Application Gateway Routing
- Demonstration – Configure an Application Gateway
- Learning Recap

Exam objectives

Configure name resolution and load balancing

- Configure an internal or public load balancer

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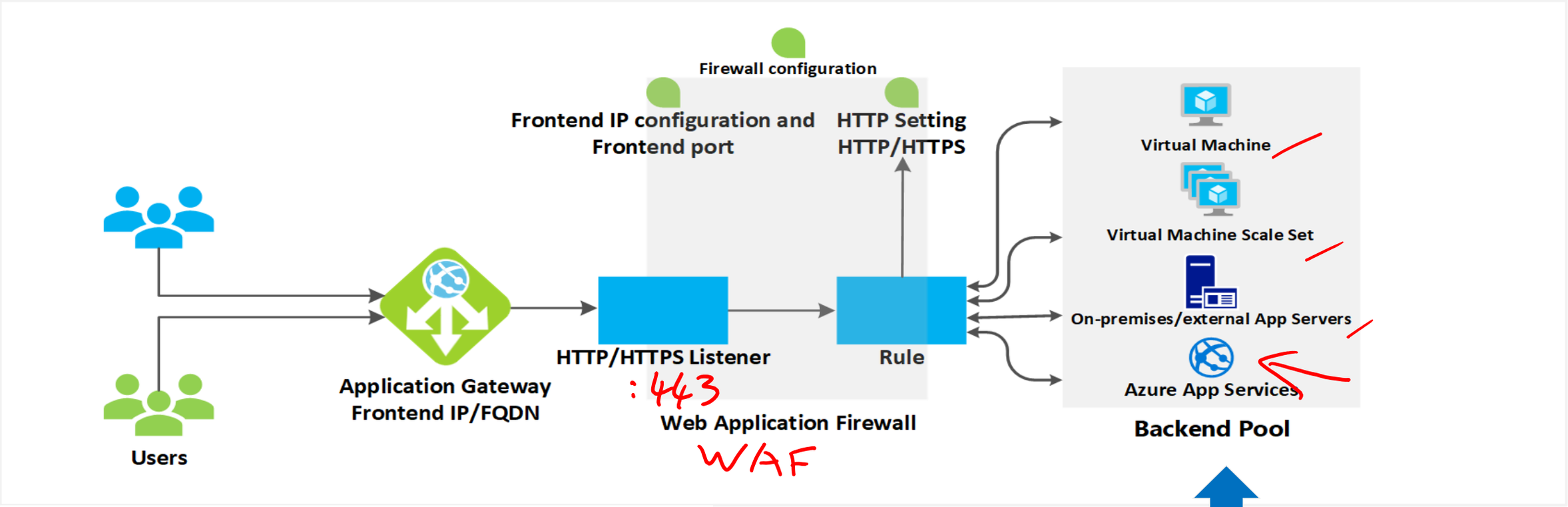
Implement Application Gateway

OWASP



- Manages web app requests and routes traffic to a backend pool based on the URL

Azure virtual machines, Azure VM scale sets, Azure App Service, and even on-premises servers

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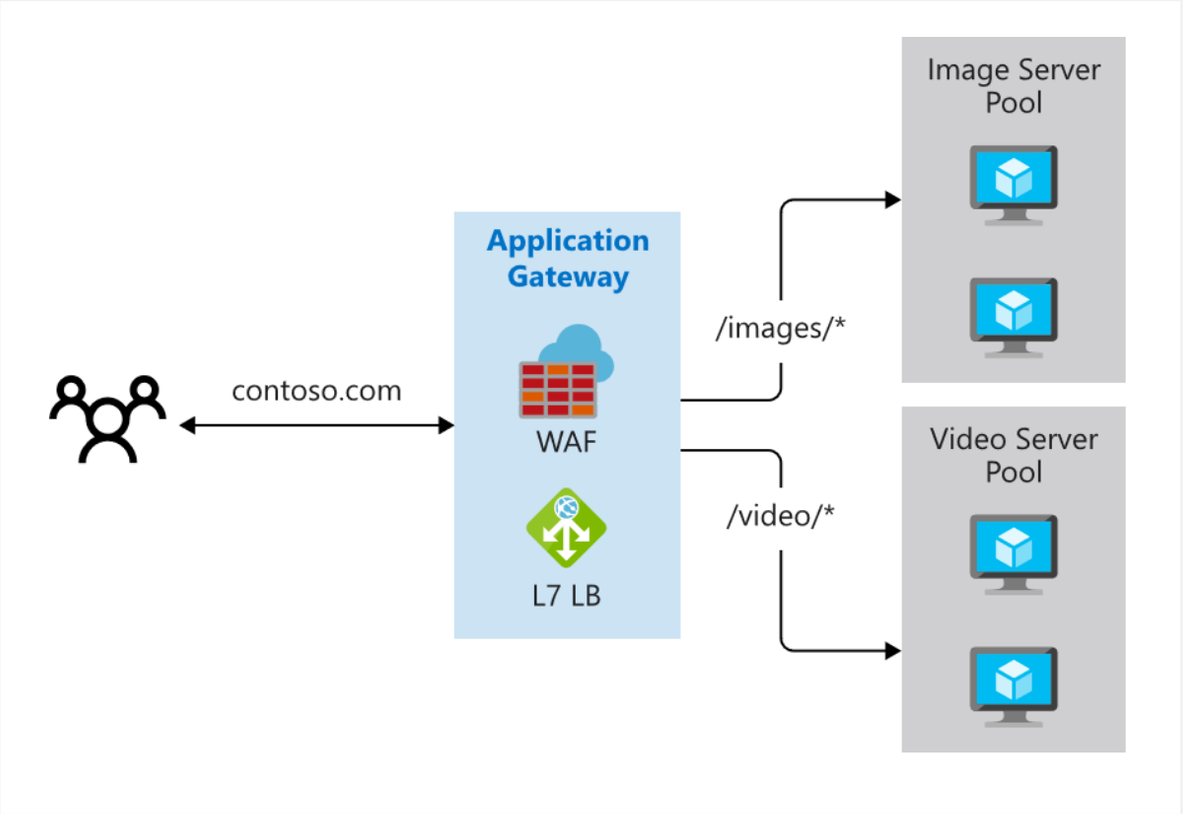
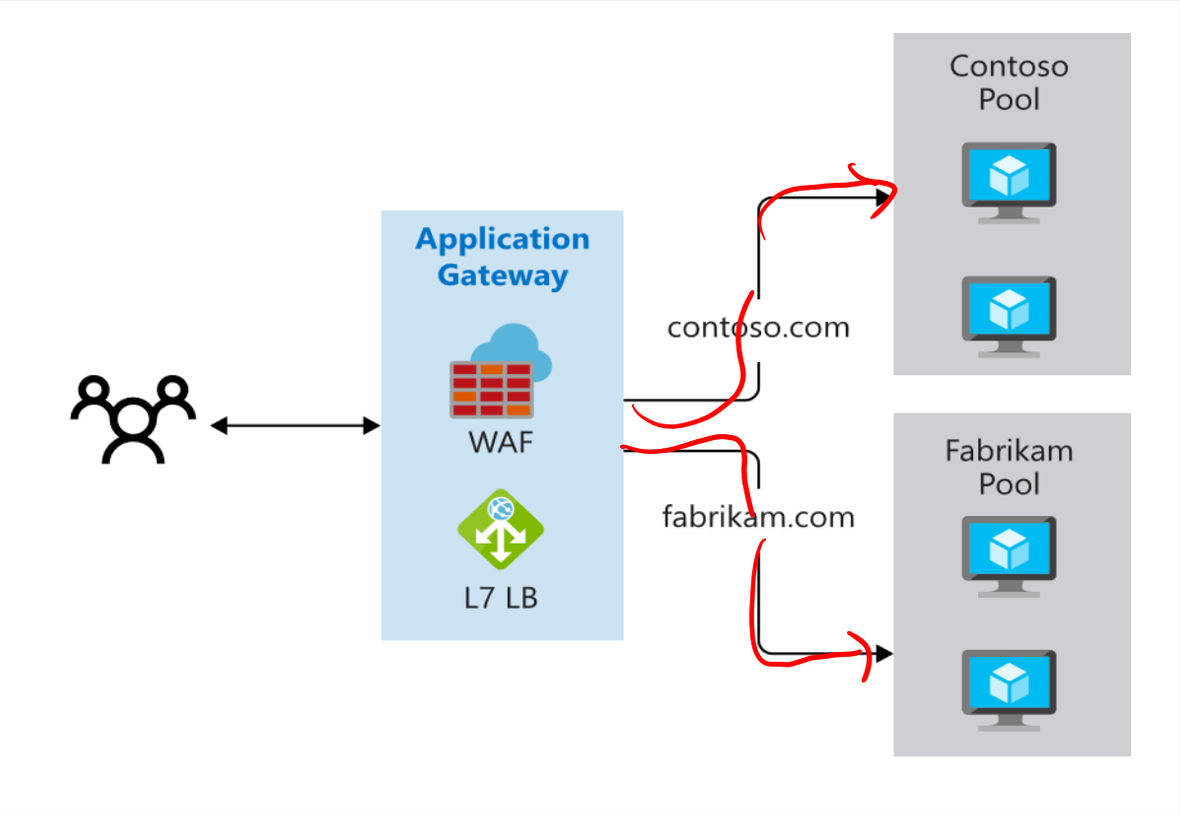
Determine Application Gateway Routing

Path-based routing



Multiple-site routing



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Introduction to Network Watcher

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Learning Objective – Introduction to Network Watcher

- What is Network Watcher?
- Understand Monitoring Tools
- Understand Traffic Tools
- Understand Network Diagnostics (3)
- Monitor and Troubleshoot Connections
- Learning Recap

Exam objectives

Monitor resources in Azure

- Use Azure Network Watcher

Configure and manage virtual networks in Azure

- Troubleshoot network connectivity

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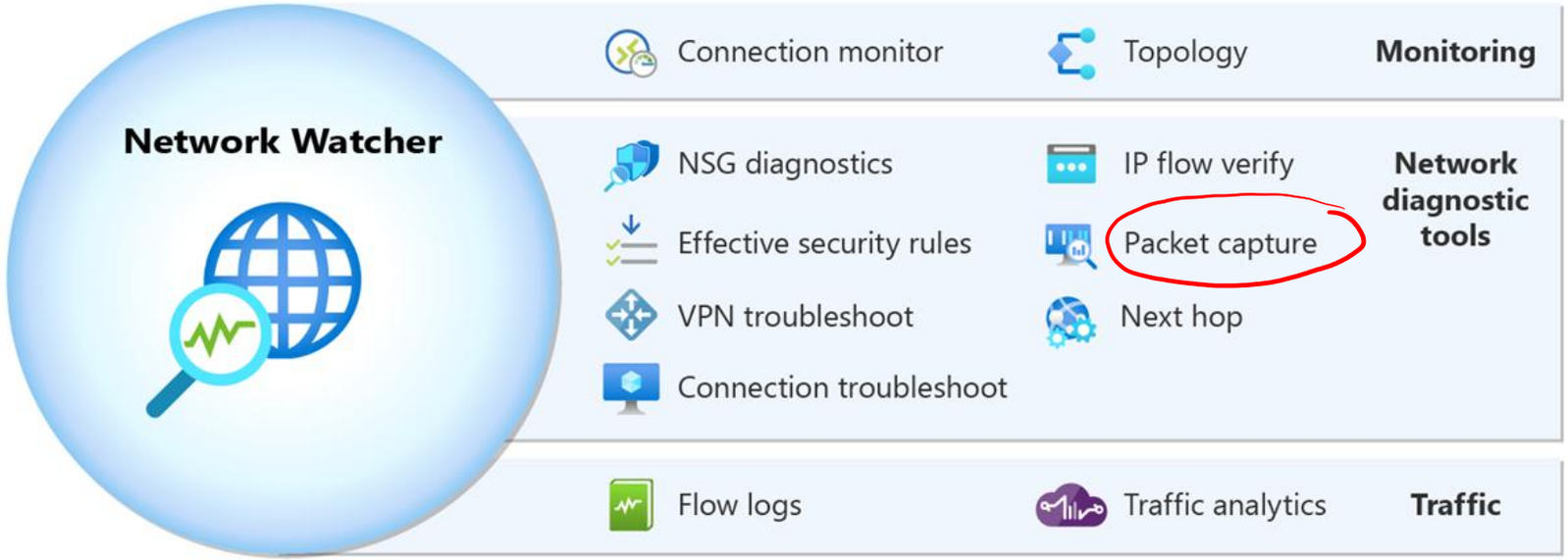
What is Network Watcher?



- Azure Network Watcher is a suite of tools
- Includes monitoring, network diagnostics, and traffic information
- Automatically enabled for every subscription.
- Can help repair the network health of IaaS products like VMs, VNets, application gateways, load balancers, etc.
- Network Watcher isn't designed or intended for PaaS monitoring or Web analytics.

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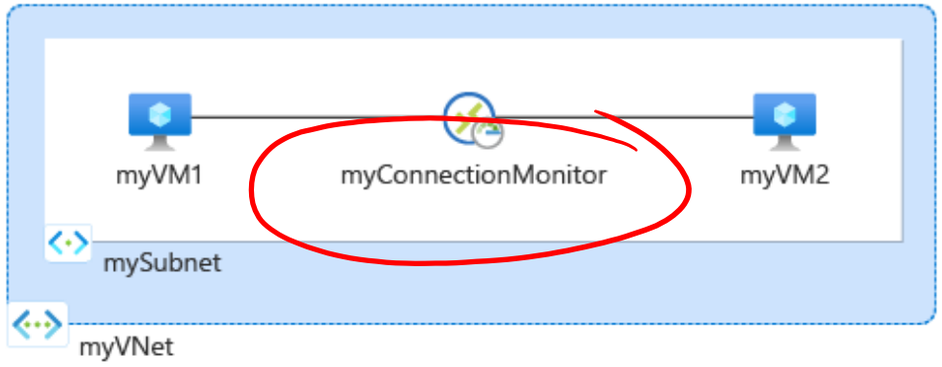
Review Monitoring Tools

Topology provides a visual representation of your networking configuration

Connection Monitor provides end-to-end connection monitoring for Azure and hybrid endpoints



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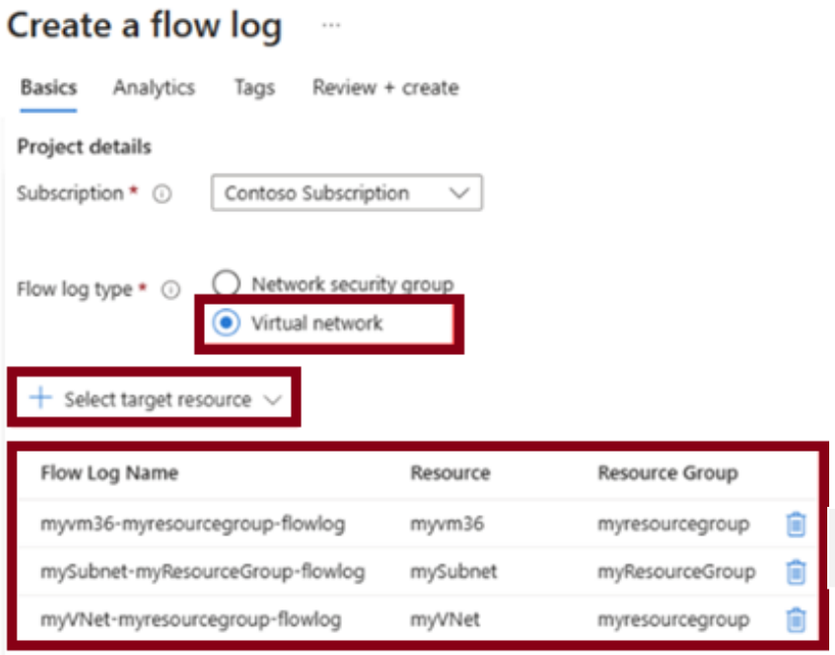
Review Traffic Tools

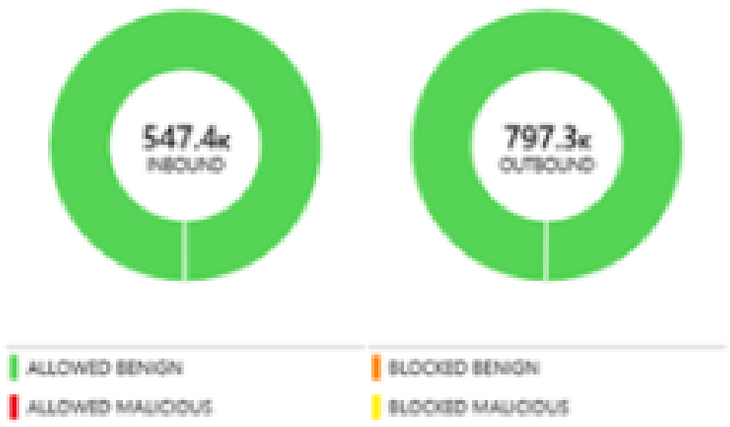
Flow Logs shows information on ingress and egress of IP traffic through a NSG or VNet



Raw Logs

Traffic analytics provides visualizations of flow logs data

- Aggregate NSG and VNet flow log records
- Enhance the records with geography, security, and topology information
- Provide analytics – traffic hotspots, traffic distribution …



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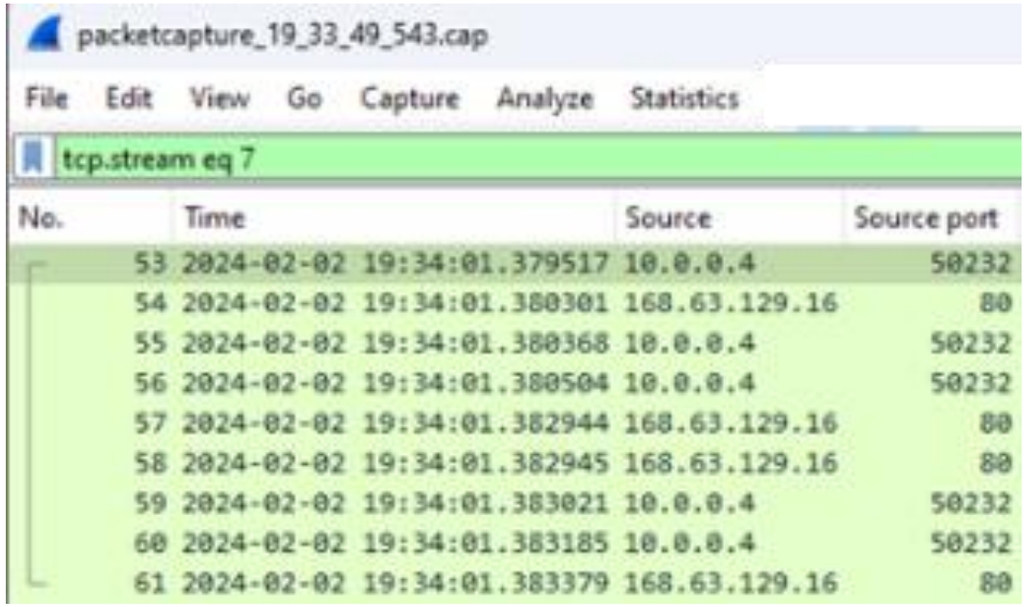
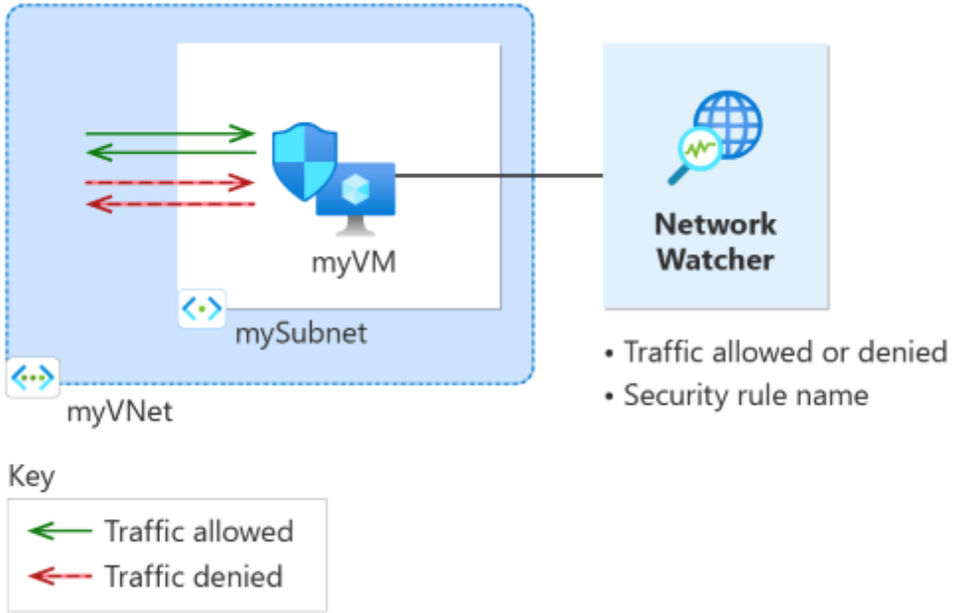
Review Network Diagnostics Tools (1 of 3)

Packet Capture creates packet capture sessions to track traffic to and from a VM or a VMSS – calculate latency, find anomalies, …



No.	Time	Source	Source port
53	2024-02-02 19:34:01.379517	10.0.0.4	50232
54	2024-02-02 19:34:01.380301	168.63.129.16	80
55	2024-02-02 19:34:01.380368	10.0.0.4	50232
56	2024-02-02 19:34:01.380504	10.0.0.4	50232
57	2024-02-02 19:34:01.382944	168.63.129.16	80
58	2024-02-02 19:34:01.382945	168.63.129.16	80
59	2024-02-02 19:34:01.383021	10.0.0.4	50232
60	2024-02-02 19:34:01.383185	10.0.0.4	50232
61	2024-02-02 19:34:01.383379	168.63.129.16	80

IP Flow Verify checks if a packet is allowed or denied to or from an Azure VM based on the configured security and admin rules.



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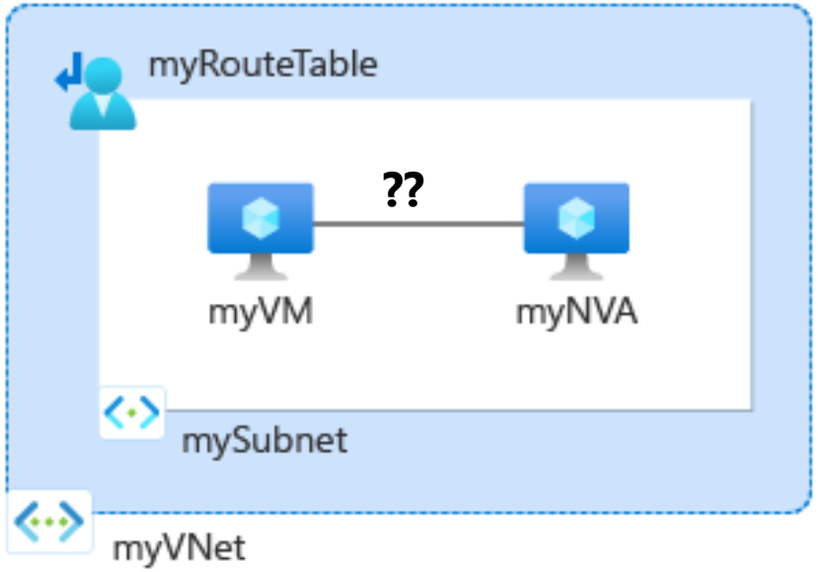
Review Network Diagnostics Tools (2 of 3)

Connection Troubleshoot helps reduce the amount of time to diagnose and troubleshoot network connectivity issues – DNS issues, missing routes …

Diagnostic tests

Test	Status	Details
Connectivity test	Unreachable	Probes sent: 30, probes failed: 30
Outbound NSG diagnostic	Allow	Outbound communication to destination is allowed
Inbound NSG diagnostic	Allow	Inbound communication to destination is allowed
Next hop (from source)	Success	Next hop type: Virtual Network Route table: System Route
Destination port accessible	Timeout	

Next Hop helps you determine if traffic is being directed to the intended destination, or whether the traffic is being dropped.



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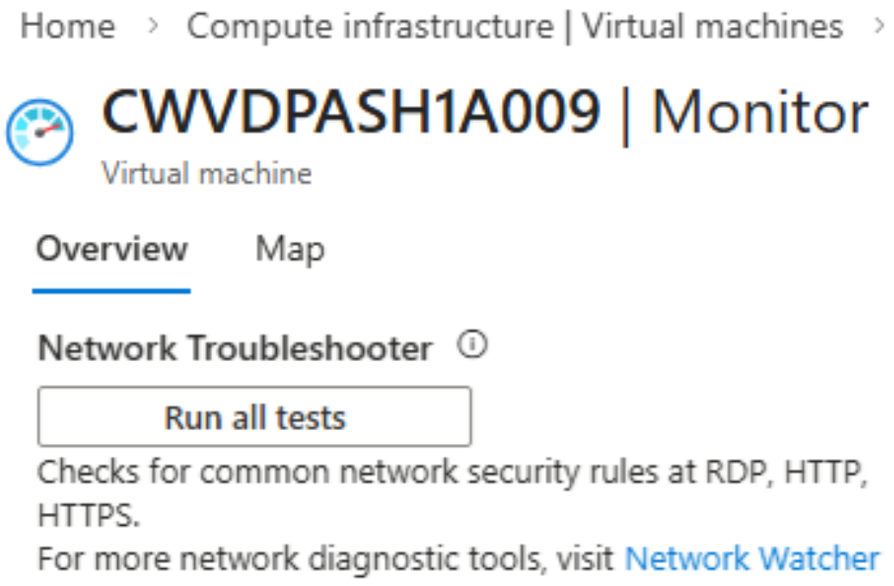
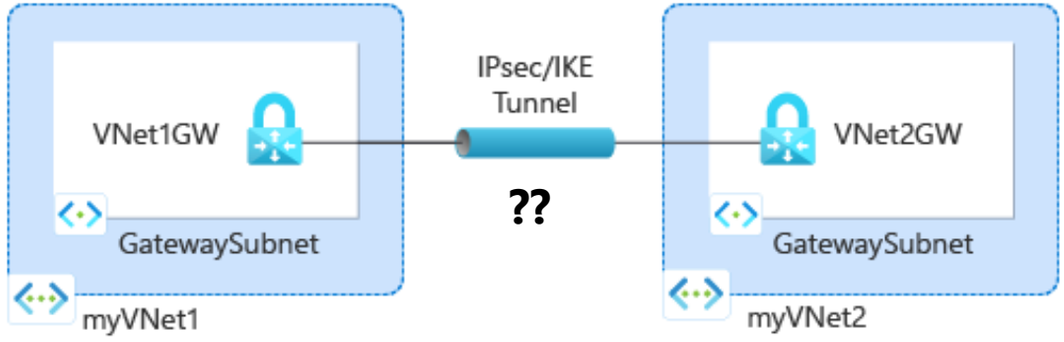
Review Network Diagnostics Tools (3 of 3)

VM Network Troubleshooter checks if commonly used ports are blocked on their Azure virtual machine.



VPN Troubleshoot helps monitor and troubleshoot virtual network gateways and their connections.



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Monitor and Troubleshoot Connections

Effective Rules view the aggregated inbound and outbound rules applied to a network interface.

Network Interface: vm01990 Effective security rules Topology

Virtual network/subnet: vnet01/subnet0 NIC Public IP: - NIC Private IP: **10.1.0.4** Accelerated networking: **Disabled**

nsg01

Inbound rules

Name	Priority	Source	Access
RDP_Inbound	100	13.23.34.45/32	Allow
DenyAllInBound	65500	0.0.0.0/0,0.0.0.0/0	Deny

Outbound rules

Name	Priority	Source	Access
AllowVnetOutBound	65000	Virtual network (1 prefixes)	Allow
DenyAllOutBound	65500	0.0.0.0/0,0.0.0.0/0	Deny

NSG Diagnostics helps you verify that your network security group rules are set up properly.

Traffic will be allowed if all NSGs allow it.

Traffic status: Denied

NSG name	Applied to	Applied action	Additional info
GlobalRules	myVNet	Allow	View details
mySubnet-nsg	mySubnet	Allow	View details
myVM-nsg	myvm36	Deny	View details

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Lab 06 – Implement Traffic Management

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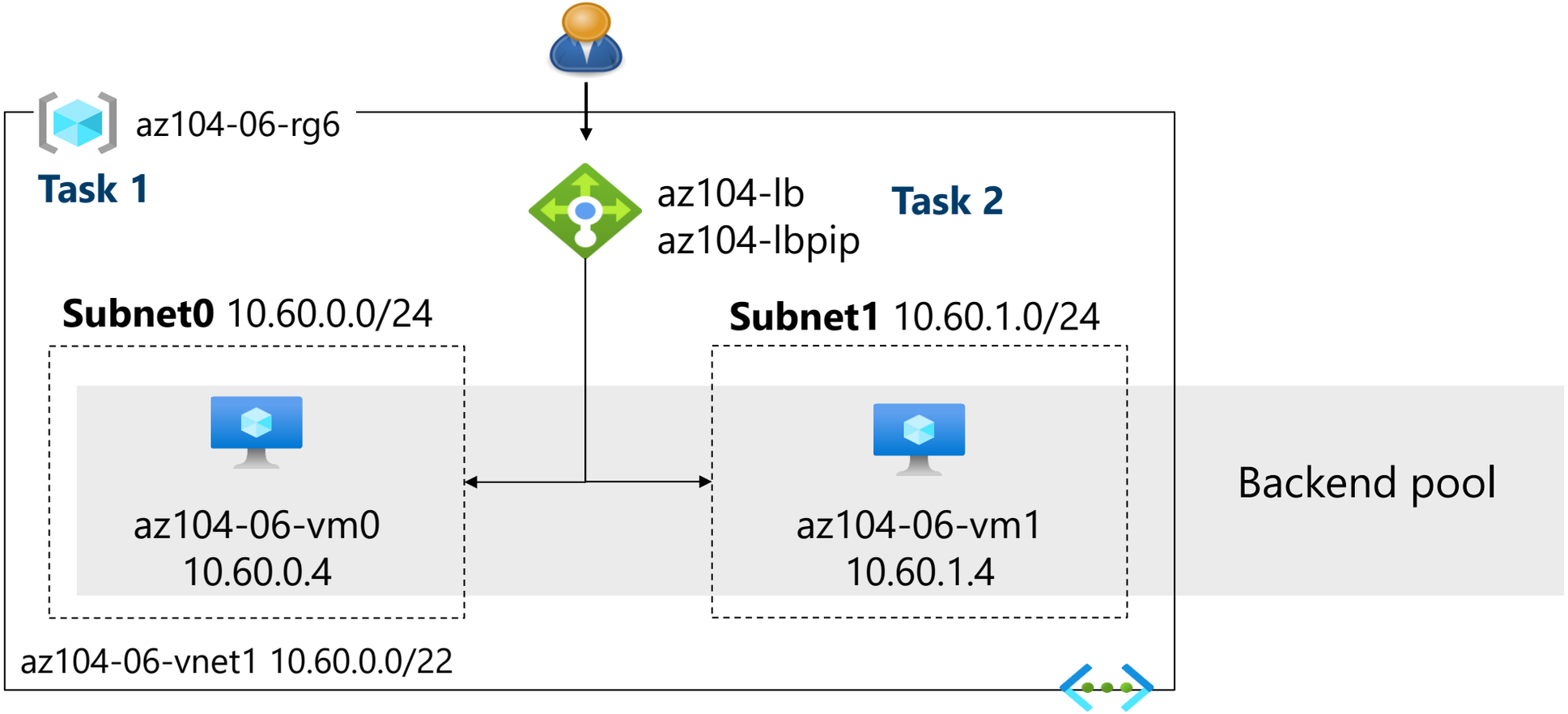
Lab 06 – Architecture Diagram (Load Balancer)



Next slide for an application gateway

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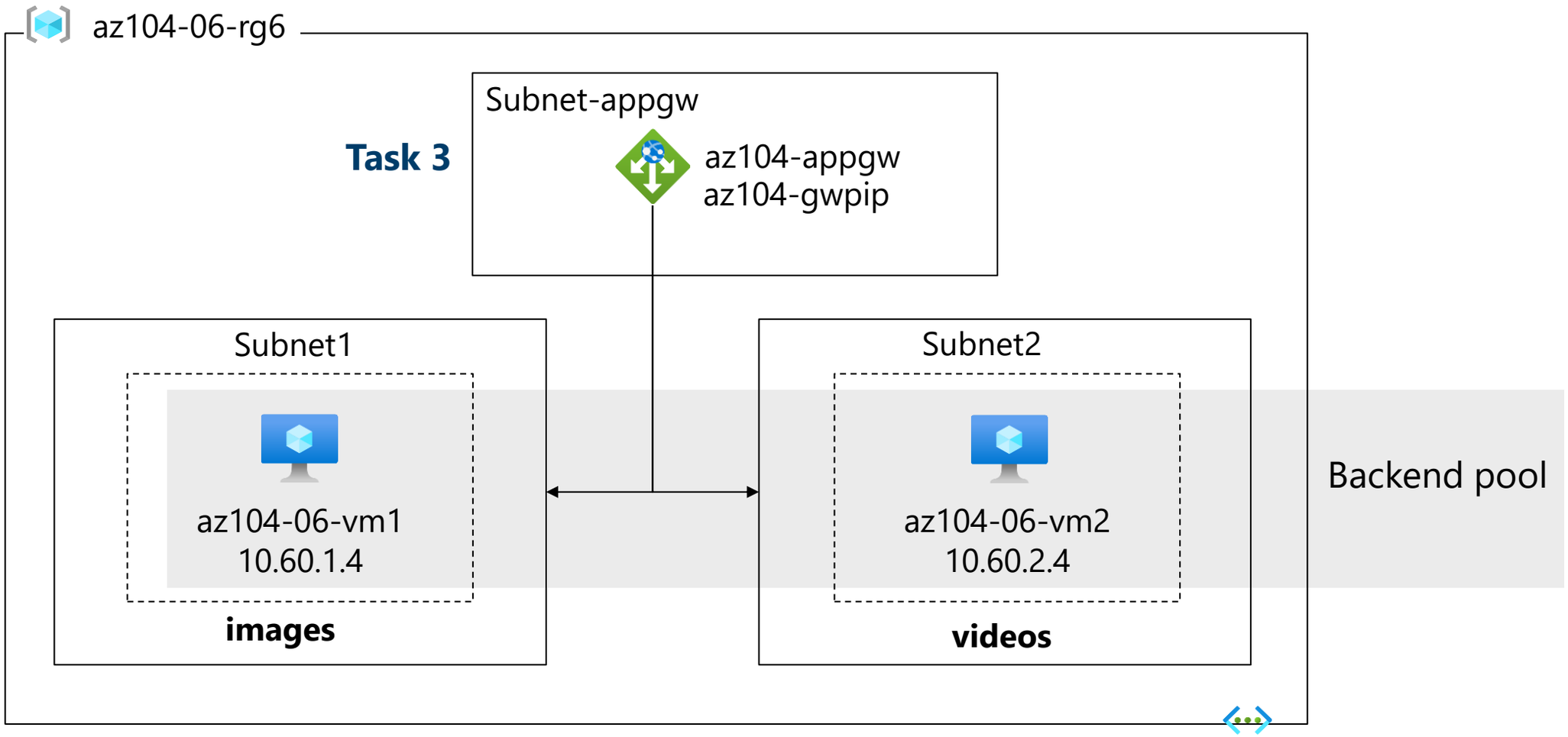
Lab 06 – Architecture Diagram (Application Gateway)



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